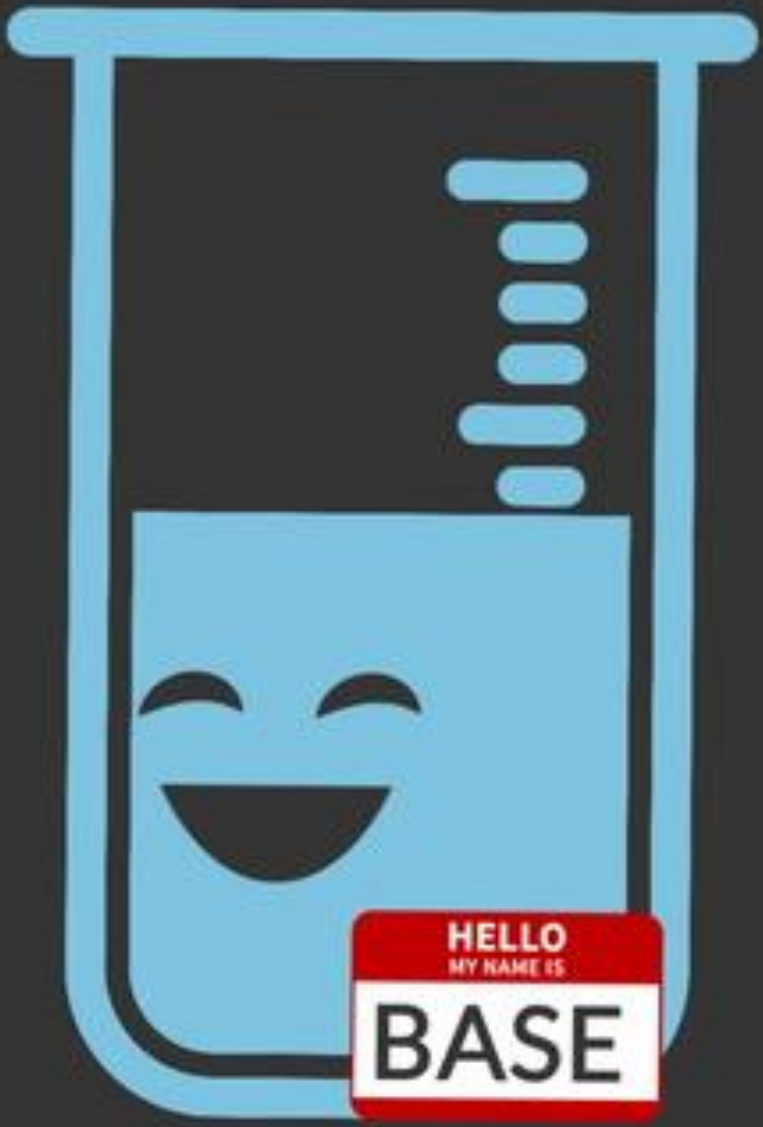


Unit 7: Acids and Bases



U6D7 AND U7D1 January 23-28 (F/W)

Learning Target: What have you learned about Electrochem?
What are Arrhenius and Bronsted-Lowry acids and bases? How do we calculate pH?

Agenda:

- ~Get ready for the U6 Mini Test!
- ~New Unit! U7: Acids and Bases- pg 1-6- listen to the first 2 lectures (Intro to A/B and pH & pOH of strong A/B)

Homework:

- ~Unit 7 Textbook Prob Set 1 going up shortly
- ~Unit 7 Exam 1 2/10-11

U7D2 January 29-30 (Th/F)

Learning Target: How is pH calculated?

Agenda:

- ~New seats?
- ~Practice- pH, pOH, K_w , naming, conjugates
- ~Quiz- pp 1-6 in packet

Homework:

- ~Unit 7 Textbook Prob Set 1 going up shortly
- ~Unit 7 Exam 1 2/10-11

U7D3 February 2-3 (M/T)

Learning Target: What is the pH of a weak acid?

Agenda:

- ~Grab a sticker! Review Exam 6
- ~Inquiry Investigation- pH and Temperature
- ~ Notes and Practice- K_a , K_b , weak acid equilibria

Homework:

- ~Unit 7 Textbook Prob Set 1 due 11:55 pm 2/9-10
- ~Unit 7 Exam 1 (of 2) 2/10-11 (Lectures 1, 2, 3.1, 3.2, 5)
- ~Unit 7 Exam 2 (of 2) 2/24-25 (Lectures 4, 6.1, 6.2, 7)

Autoionization of Water Investigation

The autoionization of water is known to be **endothermic** and may also be represented as follows:



A group of students are asked to hypothesize about the changes in K_w , the $[\text{H}_3\text{O}^+]$, and the pH of a pure water sample at temperatures COLDER than 25°C .

- 1) Write a balanced equation including a term for the heat. How does the proposed temperature change affect the equilibrium?
- 2) Draw two comparative particle diagrams of a beaker at 25°C and one at the colder temperature.
- 3) The students hypothesize that there would be no changes any of the values of K_w , the $[\text{H}_3\text{O}^+]$, and the pH. Do you agree or disagree? Justify your answer. Determine whether each would increase, decrease or stay the same.

U7D4 February 4-5 (W/Th)

Learning Target: What is a pK_a ? What is the pH of a weak base?

Agenda:

- ~Warm up! Percent Dissociation Investigation
- ~Notes and Practice- pK_a and pK_b , diprotic acids, strength of acids, K_b , weak base equilibria

Homework:

- ~Unit 7 Textbook Prob Set 1 due 11:55 pm 2/9-10
- ~Unit 7 Exam 1 (of 2) 2/10-11 (Lectures 1, 2, 3.1, 3.2, 5)
- ~Unit 7 Exam 2 (of 2) 2/24-25 (Lectures 4, 6.1, 6.2, 7)

Percent Dissociation Investigation

It is well known that the acid dissociation, K_a , is constant for a specific temperature for an acid and is therefore an intensive property (is not dependent on the amount of acid present). Can the same be said for the percent dissociation of an acid? Percent dissociation is defined as the percentage of an acidic substance that has dissociated at a specific temperature. Is the percent dissociation of an acid dependent on the concentration of the acid? If so, does it vary directly or inversely with the concentration of acid?

Complete a mini investigation considering the acid HCN ($K_a = 6.2 \times 10^{-10}$). Consider two solutions:

Solution A: 0.500 M HCN

Solution B: 0.250 M HCN

Calculate the pH and percent dissociation of both solutions and draw a conclusion about the relationship between percent dissociation and concentration of acid.

U7D5 February 6-9 (F/M)

Learning Target: What makes an acid strong or weak?

Agenda:

- ~Notes and Practice: Acidic and basic salts
- ~Review for exam 7.1

Homework:

- ~Unit 7 Textbook Prob Set 1 due 11:55 pm 2/9-10
- ~Unit 7 Exam 1 (of 2) next block (Lectures 1, 2, 3.1, 3.2, 5)
- ~Unit 7 Exam 2 (of 2) 2/24-25 (Lectures 4, 6.1, 6.2, 7)

U7D6 February 10-11 (T/W)

Learning Target: What makes a salt acidic or basic?

Agenda:

~Exam 7.1

~Notes and Practice: Acidic and Basic Salts

Homework:

~Unit 7 Textbook Prob Set 2 due 11:55 night 2/23-24

~Unit 7 Exam 2 (of 2) 2/24-25 (Lectures 4, 6.1, 6.2, 7)

U7D6 February 12-13 (Th/F)

Learning Target: What is a buffer?

Agenda:

- ~Review Exam 7.1
- ~Notes and Practice: Buffers

Homework:

- ~Unit 7 Textbook Prob Set 2 due 11:55 night 2/23-24
- ~Unit 7 Exam 2 (of 2) 2/24-25 (Lectures 4, 6.1, 6.2, 7)

U7D7 February 18-19 (W/Th)

Learning Target: How is the pH of a buffer calculated?

Agenda:

- ~Review: what is a buffer?
- ~Calculating buffer pH and the Henderson Hasselbach Equation
- ~Practice Buffer Probs WS
- ~How do we titrate a weak acid or base?

Homework:

- ~Unit 7.2 Exam moved to 2/26-27!
- ~Unit 7 Textbook Prob Set 2 due 2/25-26

U7D8 February 20-23 (F/M)

Learning Target: What is a titration curve?

Agenda:

~WARM UP: On a BIG WHITEBOARD:

1. What is the pH of a solution when 0.090 moles of HClO_2 is added to 1.0 liters of a 0.120 M solution of NaClO_2 ? The K_a of HClO_2 is 1.9×10^{-3} .
2. A student titrates 25.00 mL of HF with a 0.1500 M solution of NaOH. 36.25 mL of NaOH is required to reach the equivalence point.
 - a. Write a neutralization reaction for the titration.
 - b. Write the net ionic equation for the reaction (no spectators!).
 - c. Calculate the molarity of the acid solution.
 - d. Considering the products of the neutralization reaction, would the pH at the equivalence point be greater or less than or equal to 7? Explain.

~Notes and Practice: Titrations and Titration Curves

Homework:

~Unit 7 Textbook Prob Set 2 due 11:55 night 2/25-26

~Unit 7 Exam 2 (of 2) 2/26-27 (Lectures 4, 6.1, 6.2, 7)

~WARM UP: On a BIG WHITEBOARD:

1. What is the pH of a solution when 0.090 moles of HClO_2 is added to 1.0 liters of a 0.120 M solution of NaClO_2 ? The K_a of HClO_2 is 1.9×10^{-3} .
2. A student titrates 25.00 mL of HF with a 0.1500 M solution of NaOH. 36.25 mL of NaOH is required to reach the equivalence point.
 - a. Write a neutralization reaction for the titration.
 - b. Write the net ionic equation for the reaction (no spectators!).
 - c. Calculate the molarity of the acid solution.
 - d. Considering the products of the neutralization reaction, would the pH at the equivalence point be greater or less than or equal to 7? Explain.

U7D9/U8D1 February 24-25 (T/W)

Learning Target: What is the half equivalence point?

Agenda:

- ~Activity: Titration Curve Analysis
- ~Review: Chemical Warfare: Acid Base Style
- ~Notes and practice: Structure of an atom

Homework:

- ~Unit 7 Textbook Prob Set 2 due 11:55 night 2/25-26
- ~Unit 7 Exam 2 (of 2) next block (Lectures 4, 6.1, 6.2, 7)

U7D10 February 26-27 (Th/F)

Learning Target: What have we learned about acids and bases?

Agenda:

~Unit 7.2 Exam- you got this!

Homework:

~Relax